Byzantine Robustness Comparison up to $f = 10$: SNN Direct vs. SNN Rate (Complete Sweeps)

ByzFL SNN Benchmark

June 26, 2026

Contents

1 Introduction

This document presents side-by-side comparison grids (1x2 layout) of Byzantine robustness results for Spiking Neural Networks (SNNs) comparing Direct and Rate data encoding schemes on MNIST, extended up to $f = 10$ Byzantine clients. The configurations compared are:

1. **SNN Direct**: Spiking Neural Network with Direct encoding on MNIST (‘plots_direct_vs_rate_f10/snn_direct‘)

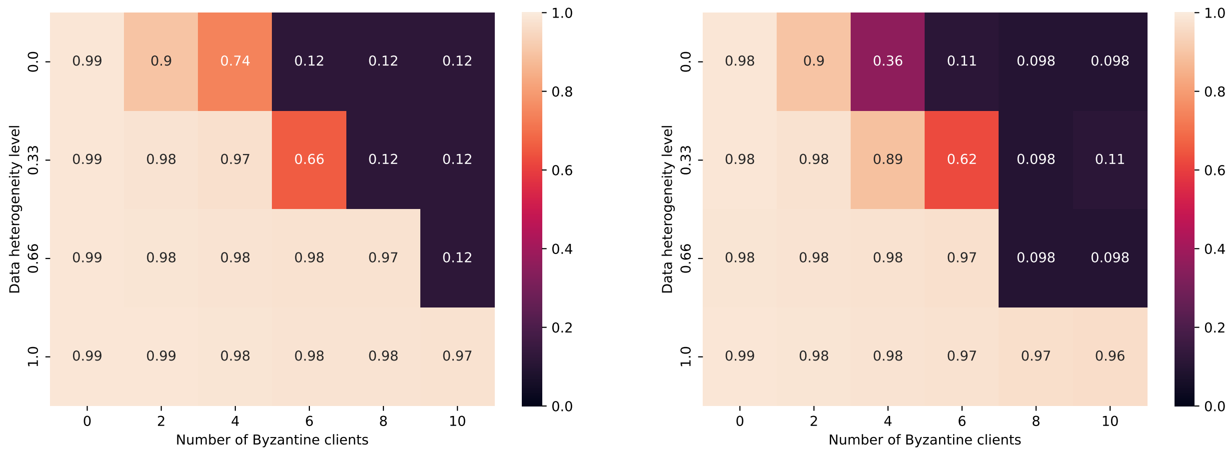
2. **SNN Rate**: Spiking Neural Network with Rate encoding on MNIST (‘plots_direct_vs_rate_f10/snn_rate‘)

Each grid illustrates the test accuracy under varying numbers of Byzantine clients ($f \in [0, 2, 4, 6, 8, 10]$) and data heterogeneity levels ($\gamma \in [0.0, 0.33, 0.66, 1.0]$) using a single training seed (seed 42), evaluated at the validation-based best step.

2 Best Overall (Worst-Case Across All Attacks)

This section compares performance under the worst-case attack scenario (minimizing the maximum damage over all attacks) for the dynamically selected best aggregator, and for each of the fixed aggregators.

2.1 Best Aggregator Choice

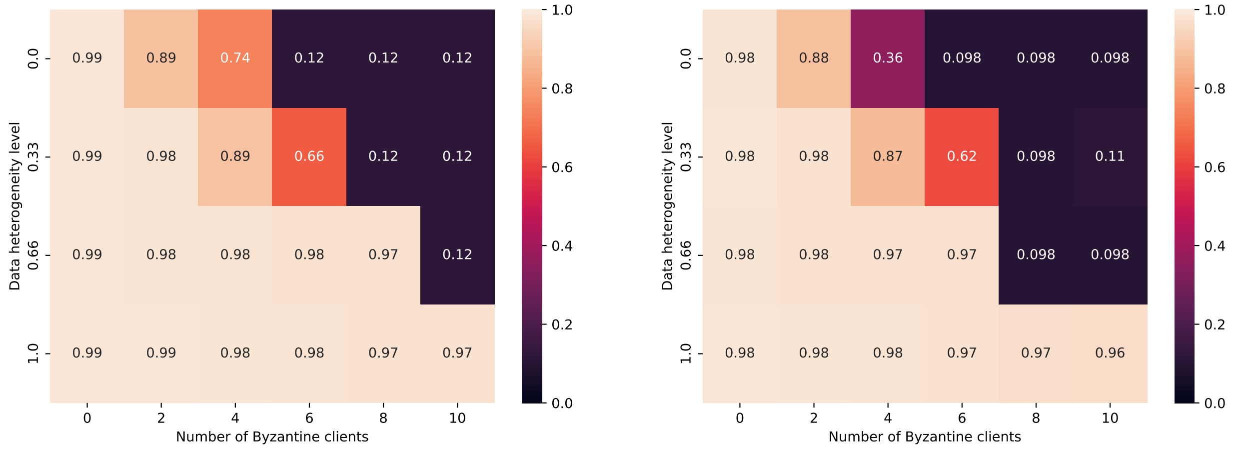


(a) SNN Direct

(b) SNN Rate

Figure 1: Worst-case test accuracy comparison using the best overall aggregator selection evaluated at the validation-based best step.

2.2 Centered Clipping

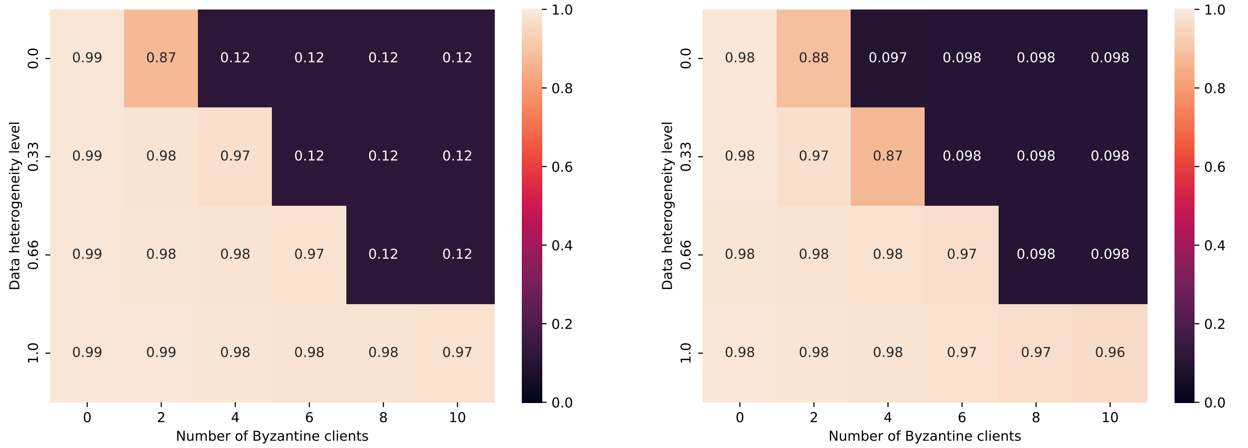


(a) SNN Direct

(b) SNN Rate

Figure 2: Worst-case test accuracy comparison using Centered Clipping.

2.3 Geometric Median

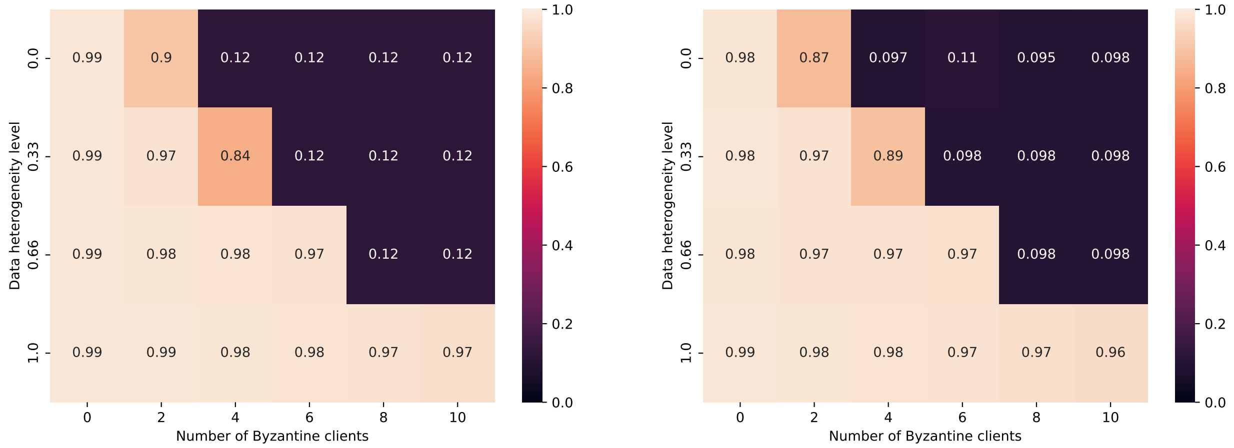


(a) SNN Direct

(b) SNN Rate

Figure 3: Worst-case test accuracy comparison using Geometric Median.

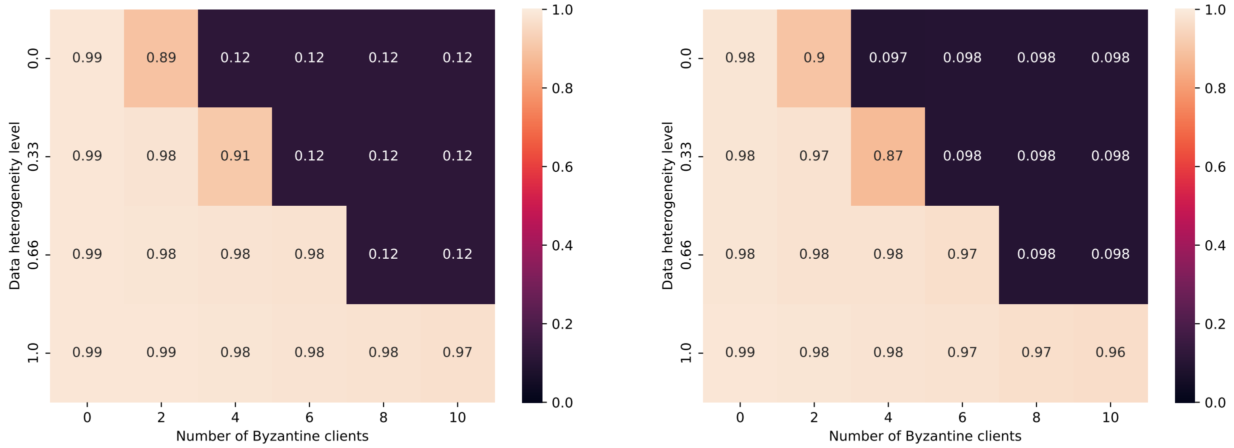
2.4 Multi-Krum



(a) SNN Direct (b) SNN Rate

Figure 4: Worst-case test accuracy comparison using Multi-Krum.

2.5 Trimmed Mean



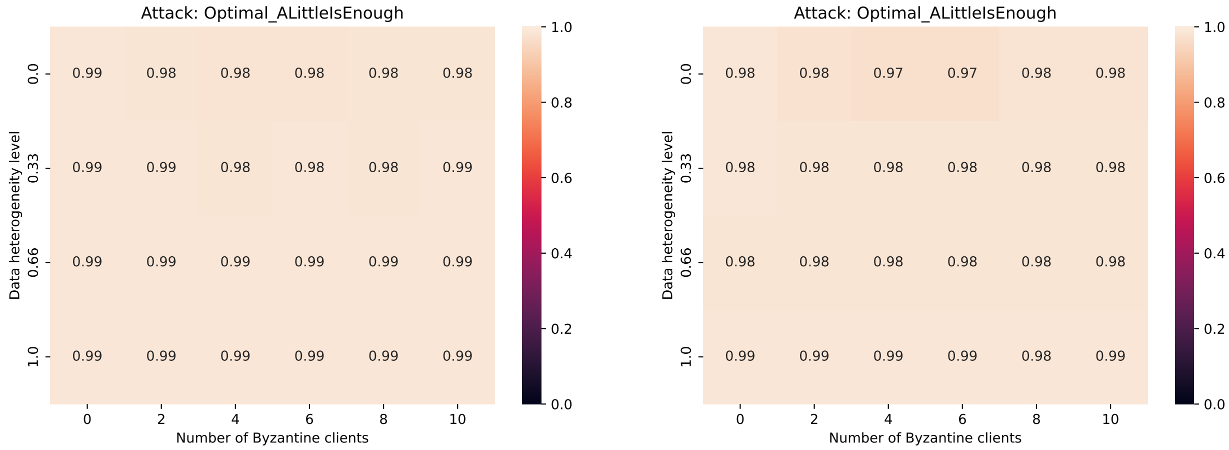
(a) SNN Direct (b) SNN Rate

Figure 5: Worst-case test accuracy comparison using Trimmed Mean.

3 Fixed Attack: Optimal A Little Is Enough (ALIE)

This section compares performance under the original Optimal ALIE Byzantine attack.

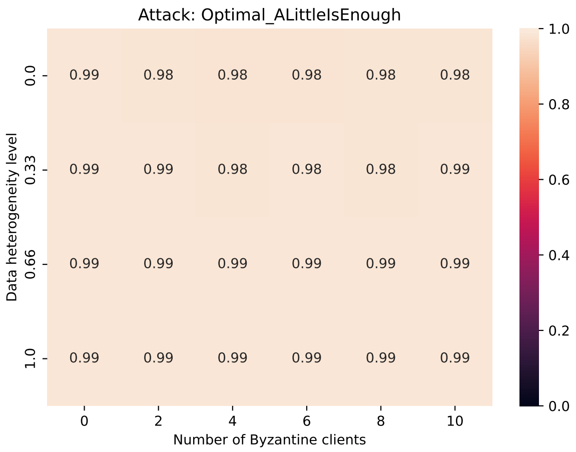
3.1 Best Aggregator Choice



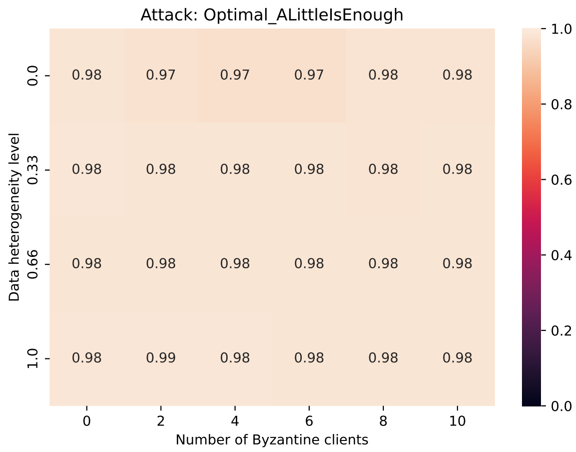
(a) SNN Direct (b) SNN Rate

Figure 6: Accuracy comparison under Optimal ALIE attack using the best aggregator selection evaluated at the validation-based best step.

3.2 Centered Clipping



(a) SNN Direct

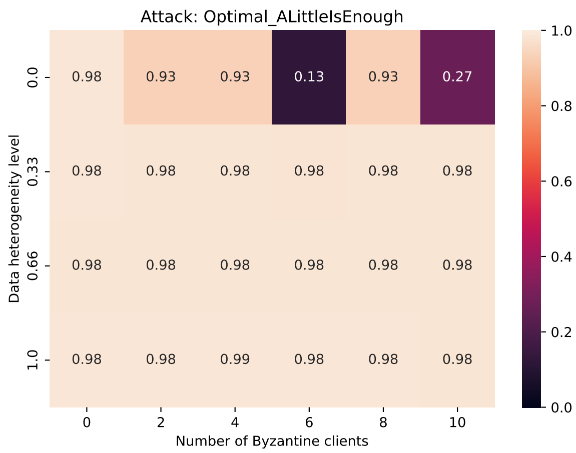


(b) SNN Rate

Figure 7: Accuracy comparison under Optimal ALIE attack using Centered Clipping.

3.3 Geometric Median

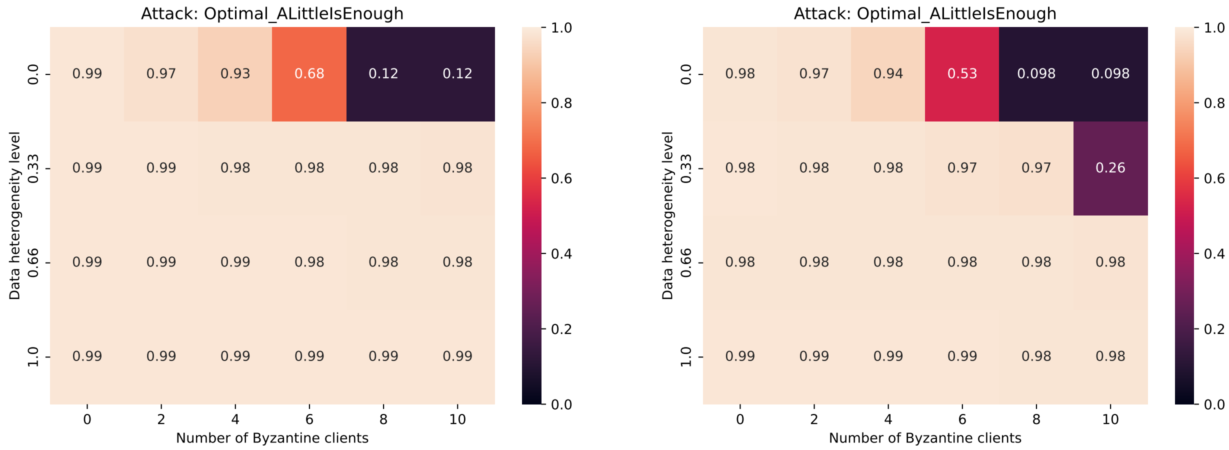
(a) SNN Direct



(b) SNN Rate

Figure 8: Accuracy comparison under Optimal ALIE attack using Geometric Median.

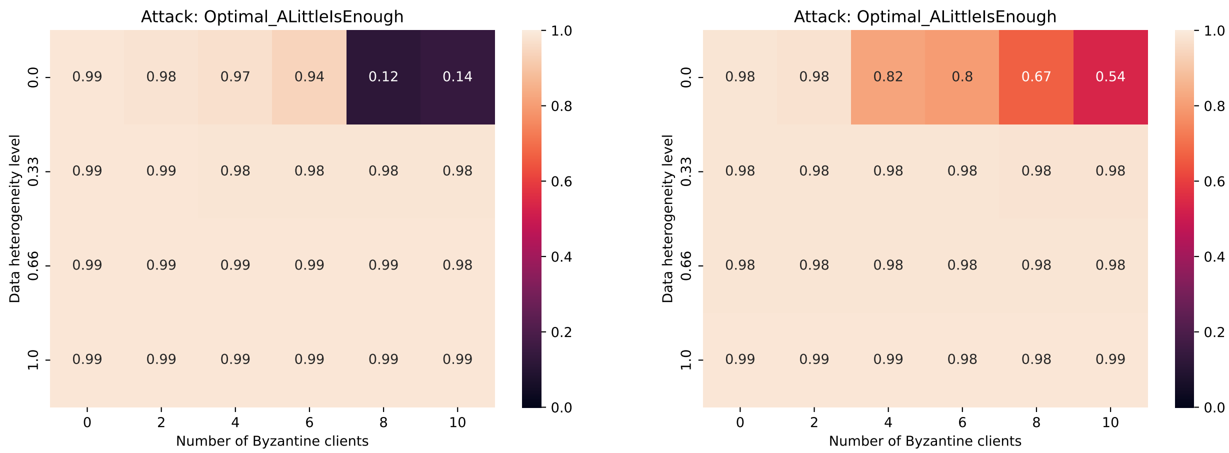
3.4 Multi-Krum



(a) SNN Direct (b) SNN Rate

Figure 9: Accuracy comparison under Optimal ALIE attack using Multi-Krum.

3.5 Trimmed Mean



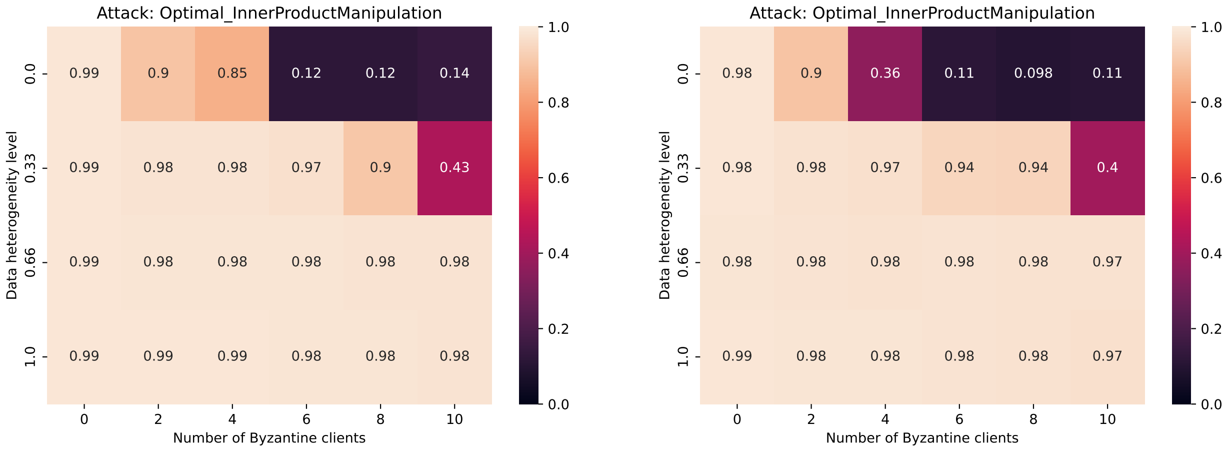
(a) SNN Direct (b) SNN Rate

Figure 10: Accuracy comparison under Optimal ALIE attack using Trimmed Mean.

4 Fixed Attack: Optimal Inner Product Manipulation (IPM)

This section compares performance under the Optimal IPM Byzantine attack.

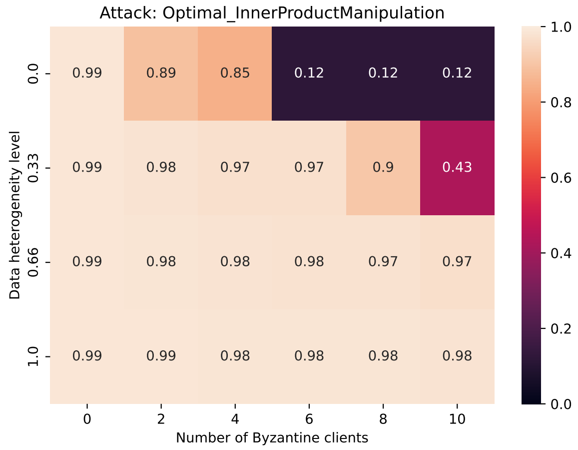
4.1 Best Aggregator Choice



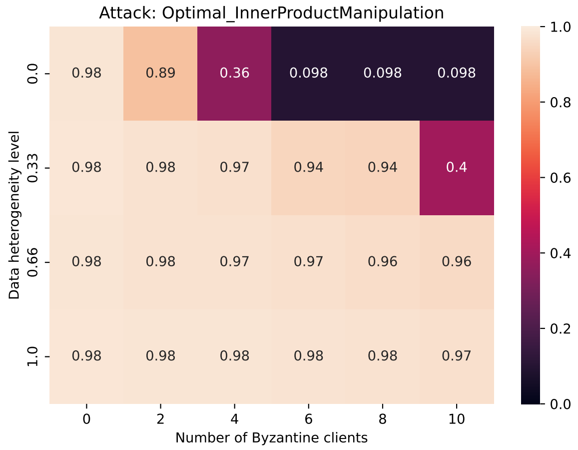
(a) SNN Direct (b) SNN Rate

Figure 11: Accuracy comparison under Optimal IPM attack using the best aggregator selection evaluated at the validation-based best step.

4.2 Centered Clipping



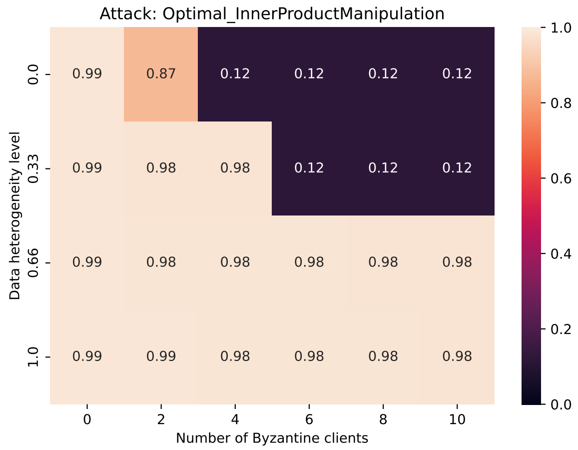
(a) SNN Direct



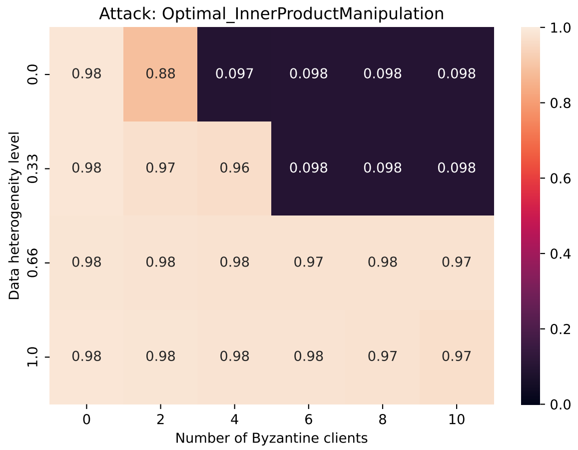
(b) SNN Rate

Figure 12: Accuracy comparison under Optimal IPM attack using Centered Clipping.

4.3 Geometric Median



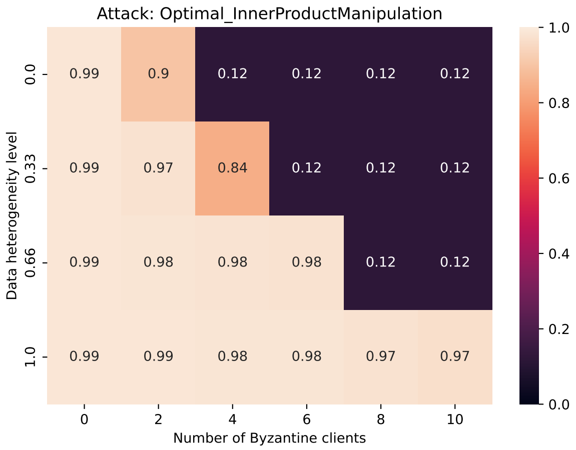
(a) SNN Direct



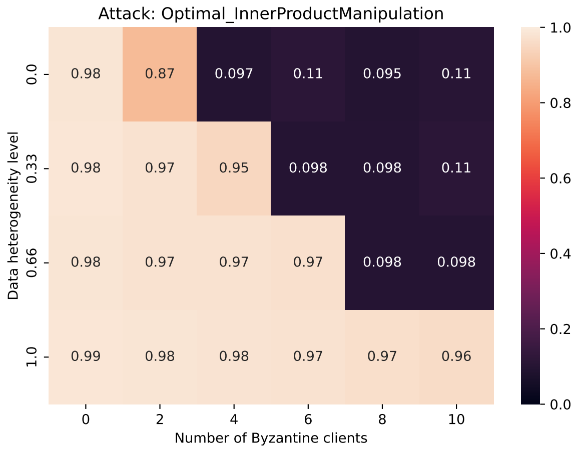
(b) SNN Rate

Figure 13: Accuracy comparison under Optimal IPM attack using Geometric Median.

4.4 Multi-Krum



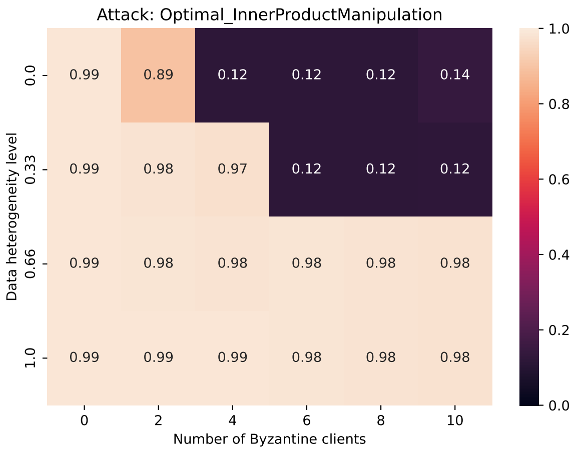
(a) SNN Direct



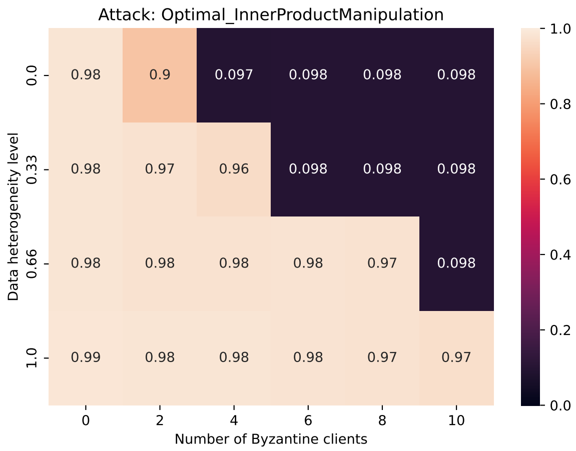
(b) SNN Rate

Figure 14: Accuracy comparison under Optimal IPM attack using Multi-Krum.

4.5 Trimmed Mean



(a) SNN Direct



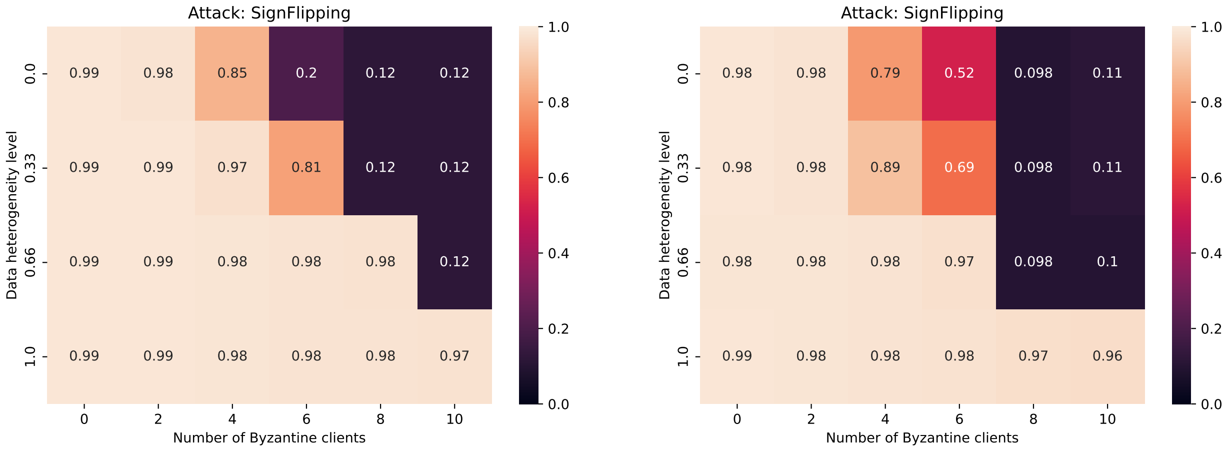
(b) SNN Rate

Figure 15: Accuracy comparison under Optimal IPM attack using Trimmed Mean.

5 Fixed Attack: Sign Flipping

This section compares performance under the Sign Flipping Byzantine attack.

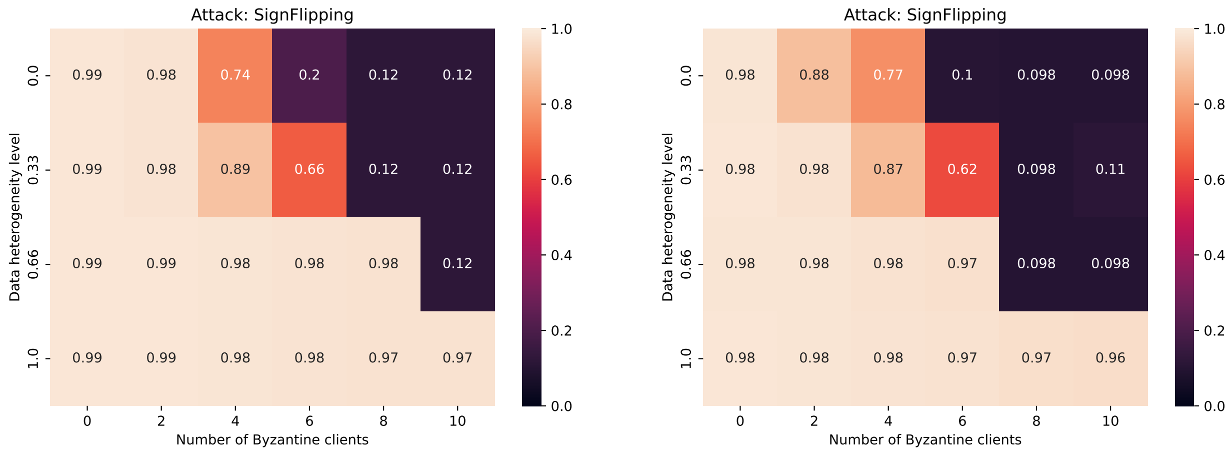
5.1 Best Aggregator Choice



(a) SNN Direct (b) SNN Rate

Figure 16: Accuracy comparison under Sign Flipping attack using the best aggregator selection evaluated at the validation-based best step.

5.2 Centered Clipping

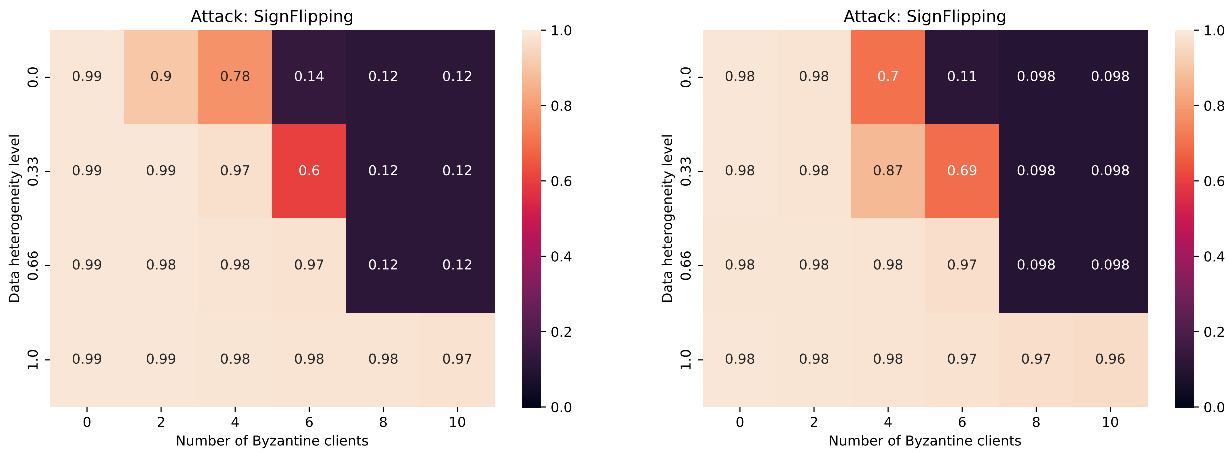


(a) SNN Direct

(b) SNN Rate

Figure 17: Accuracy comparison under Sign Flipping attack using Centered Clipping.

5.3 Geometric Median

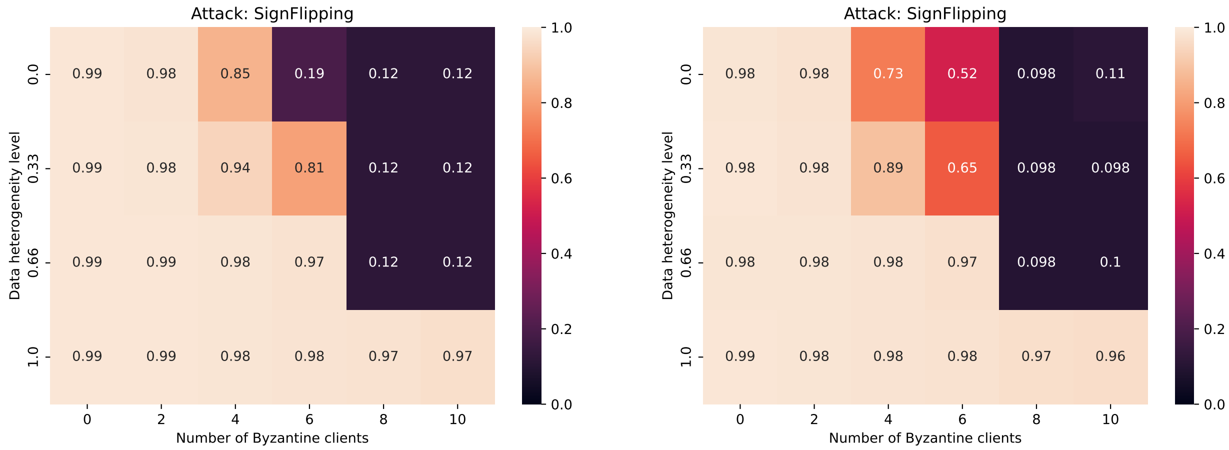


(a) SNN Direct

(b) SNN Rate

Figure 18: Accuracy comparison under Sign Flipping attack using Geometric Median.

5.4 Multi-Krum

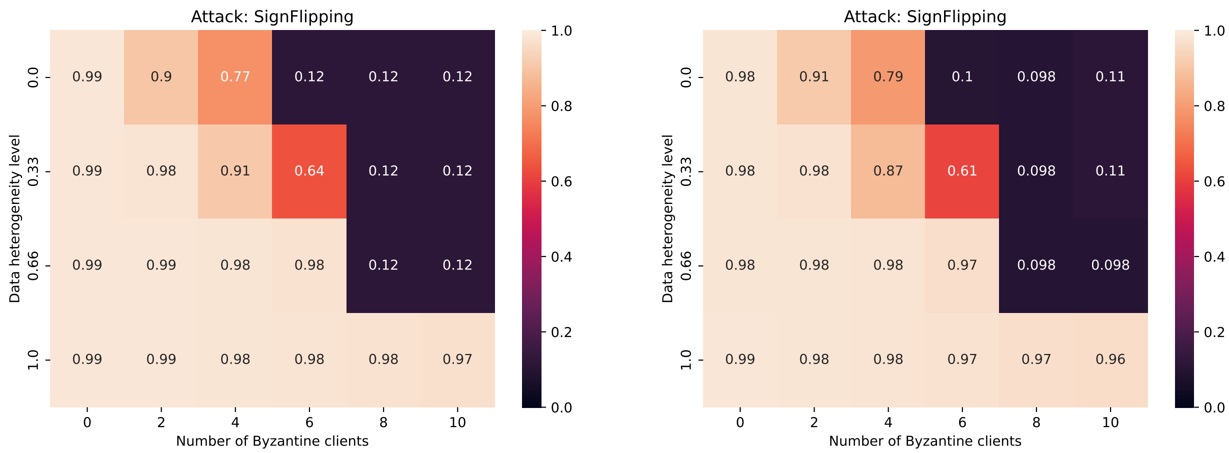


(a) SNN Direct

(b) SNN Rate

Figure 19: Accuracy comparison under Sign Flipping attack using Multi-Krum.

5.5 Trimmed Mean



(a) SNN Direct

(b) SNN Rate

Figure 20: Accuracy comparison under Sign Flipping attack using Trimmed Mean.